

Problem

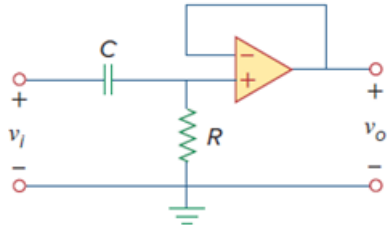
The filter in Fig. 14.90(b) has a 3-dB cutoff frequency at 1 kHz. If its input is connected to a 120-mV variable frequency signal, find the output voltage at:

(a) 200 Hz

(b) 2 kHz

(c) 10 kHz

Figure 14.90:



Step-by-step solution

Step 1 of 6

Refer to Figure 14.90(b) in the textbook.

Observe that the output is taken across the resistance in the circuit since there is no feedback loop connected to the ideal op-amp.

Use voltage division method to find the expression of the output voltage.

$$\begin{aligned} \mathbf{V}_o(s) &= \left(\frac{R}{R + \frac{1}{sC}} \right) \mathbf{V}_i(s) \\ &= \left(\frac{R}{\frac{sRC + 1}{sC}} \right) \mathbf{V}_i(s) \\ &= \left(\frac{sRC}{sRC + 1} \right) \mathbf{V}_i(s) \end{aligned}$$

Replace s with $j\omega$.

$$\mathbf{V}_o(j\omega) = \left(\frac{j\omega RC}{j\omega RC + 1} \right) \mathbf{V}_i(j\omega)$$

This is the output function of a high-pass filter.

Step 2 of 6

Simplify the expression further.

$$\begin{aligned}\mathbf{V}_o(j\omega) &= \left(\frac{1}{\frac{j\omega RC + 1}{j\omega RC}} \right) \mathbf{V}_i(j\omega) \\ &= \left(\frac{1}{1 - \frac{j}{\omega RC}} \right) \mathbf{V}_i(j\omega)\end{aligned}$$

Here, the product RC is the time constant of the circuit.

$$\tau = RC$$

Therefore, the cut-off frequency is,

$$\begin{aligned}\omega_0 &= \frac{1}{\tau} \\ &= \frac{1}{RC}\end{aligned}$$

Replace the expression of cut-off frequency in the equation of $\mathbf{V}_o(j\omega)$.

$$\begin{aligned}\mathbf{V}_o(j\omega) &= \left(\frac{1}{1 - \frac{j\omega_0}{\omega}} \right) \mathbf{V}_i(j\omega) \\ \mathbf{V}_o(jf) &= \frac{\mathbf{V}_i(jf)}{1 - \frac{jf_0}{f}}\end{aligned}$$

Step 3 of 6

Consider that the 3-dB cut-off frequency of the circuit is 1 kHz.

Substitute the 1 kHz for f_0 in the equation of $\mathbf{V}_o(j\omega)$.

$$\mathbf{V}_o(jf) = \frac{\mathbf{V}_i(jf)}{1 - \frac{j1000}{f}}$$

The magnitude of output voltage is,

$$\begin{aligned}|\mathbf{V}_o(jf)| &= \left| \frac{\mathbf{V}_i(jf)}{1 - \frac{j1000}{f}} \right| \\ &= \frac{|\mathbf{V}_i(jf)|}{\sqrt{1 + \frac{1000^2}{f^2}}}\end{aligned}$$

Consider that the magnitude value of input voltage is 120 mV. Therefore,

$$|\mathbf{V}_o(jf)| = \frac{120 \text{ mV}}{\sqrt{1 + \frac{10^6}{f^2}}}$$

Step 4 of 6

(a)

Consider that the frequency, $f = 200$ Hz.

Find the magnitude of the output voltage at 200 Hz.

$$\begin{aligned} |V_o(j200)| &= \frac{0.12}{\sqrt{1 - \frac{10^6}{200^2}}} \\ &= \frac{0.12}{\sqrt{1 + 25}} \\ &= \frac{0.12}{5.099} \\ &= 23.53 \text{ mV} \end{aligned}$$

Thus, the output voltage at 200 Hz is 23.53 mV.

Step 5 of 6

(b)

Consider that the frequency, $f = 2$ kHz.

Find the magnitude of the output voltage at 2 kHz.

$$\begin{aligned} |V_o(j2000)| &= \frac{0.12}{\sqrt{1 - \frac{10^6}{2000^2}}} \\ &= \frac{0.12}{\sqrt{1 + 0.25}} \\ &= \frac{0.12}{1.118} \\ &= 107.3 \text{ mV} \end{aligned}$$

Thus, the output voltage at 2 kHz is 107.3 mV.

Step 6 of 6

(c)

Consider that the frequency, $f = 10$ kHz.

Find the magnitude of the output voltage at 10 kHz.

$$\begin{aligned} |V_o(j10000)| &= \frac{0.12}{\sqrt{1 - \frac{1000^2}{10000^2}}} \\ &= \frac{0.12}{\sqrt{1 + 0.01}} \\ &= \frac{0.12}{1.005} \\ &= 119.4 \text{ mV} \end{aligned}$$

Thus, the output voltage at 10 kHz is 119.4 mV.